priority queue always return the minimum value contained. With Bloom filters, we discovered a new category, Monte Carlo algorithms, that might return sub-optimal solutions, and that might even return 2 different values running twice on the same input/context.

- **ADT vs CDT**: We can improve our code's performance over several levels. We can choose the best *abstract data structure* to perform certain operations, but we can also choose among different implementations, or *concrete data structures*, of an ADT, and that choice also makes a difference.
- **Dictionary**: many common problems in computer science revolve around keeping track of values; it might be URLs browsed by a crawler, documents examined by an indexer, or values to store in a cache, just to name a few. All these use cases can make use of an abstract data type called *set*, a specialization of another a
  ADT, the *dictionary*.
- **Set**: Depending on the context, there are different additional constraints for the implementation of a *set* that we might want to use. Memory is usually the greatest concern:
  - When the number of keys we actually need to store at any given time is small with respect to the size of the domain (all the possible keys), then we can use hashing to save memory.
  - When the volume of keys to store needs to scale up (but still sensibly smaller than the number of possible keys), and the amount of information we